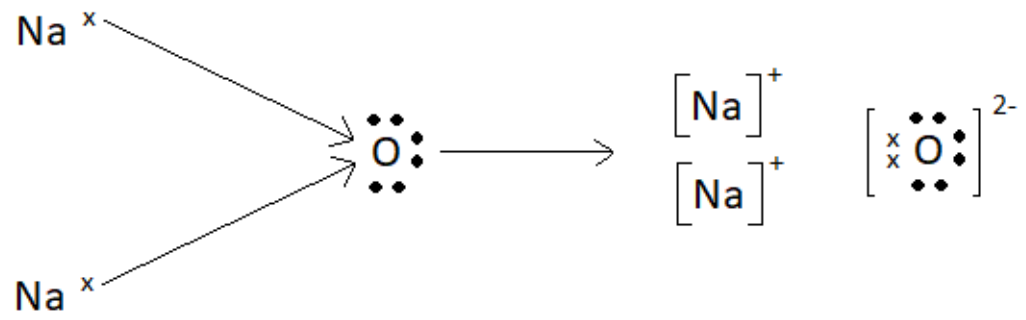
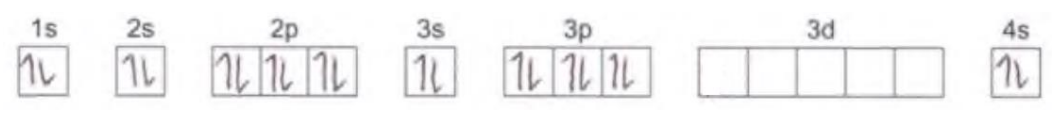


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				transfer of electrons / correct electronic structure of ions (1) charges on ions (1) 		2		2		
2.				(very) electronegative nitrogen / oxygen / fluorine both needed (accept correct compound e.g. H ₂ O)	1			1		
3.					1			1		
4.				47.1%		1		1	1	
5.				a covalent bond where the shared pair of electrons comes from the same atom	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)			^{231}Th		1		1		
	(b)			$1/16^{\text{th}}$		1		1	1	
7.				$n(\text{SO}_2) = \frac{9.1}{64.1} = 0.142 \quad (1)$ $0.142 \times 40.1 = 5.69 / 5.7 \text{ g} \quad (1)$		2		2	1	
				Section A total	3	7	0	10	3	0